# Beyond Text: Bridging Speech and Satellite Imagery for Efficient Visual Grounding

Yaoyong Liu, Yue Zhou, *Member, IEEE*, and Xue Yang, *Member, IEEE*

***Abstract*—While remote sensing visual grounding (RSVG) has emerged as a crucial bridge for human–machine interaction in Earth observation, existing methods rely exclusively on textual queries. In time-critical applications such as disaster rescue and emergency response, this reliance restricts hands-free operation and strips away vital prosodic cues like tone and emphasis, which are highly effective for disambiguating complex geospatial targets. To overcome these limitations, we formally introduce the task of audio-driven RSVG (Audio-RSVG) and construct AudioDetectIon in optical remote sensing (DIOR), a large-scale dataset simulating real-world acoustic environments with diverse timbres and background noise conditions. Furthermore, we propose RSTalker, a robust end-to-end framework designed to directly align speech signals with aerial imagery. By bypassing cascaded automatic speech recognition (ASR) pipelines, RSTalker prevents transcription error propagation, particularly regarding remote sensing terminology. This approach also effectively exploits rich acoustic semantics that are often lost during intermediate text generation. Extensive experiments demonstrate that RSTalker achieves state-of-the-art localization accuracy and noise robustness. Notably, by eliminating manual data entry and heavy decoder stacks, the proposed paradigm delivers a 3.5× speedup in complete interaction-to-inference time compared to conventional text-based baselines. This paves the way for intuitive, hands-free interaction in urgent remote sensing tasks.**

***Index Terms*—Audio-based grounding, multimodal, remote sensing, visual grounding.**

## I. Introduction

With the rapid advancement of Earth observation technologies [1], [2], [3], the volume and spatial resolution of remote sensing imagery have grown exponentially. Efficiently and accurately interpreting this massive data stream is critical for time-sensitive and high-stakes applications, such as disaster rescue, traffic control, and military surveillance [4], [5]. In these operational environments, users must quickly pinpoint specific targets of interest within vast geographical areas. Consequently, remote sensing visual grounding (RSVG) [6], [7], [8], [9], [10] has emerged as a pivotal task, serving as a vital bridge between human intent and visual perception. By localizing objects described by natural language queries (e.g., "find the red ship entering the harbor"), RSVG enables intelligent and intuitive interaction with complex geospatial data. However, unlike standard computer vision tasks, RSVG faces unique domain-specific challenges, including extreme scale variations, arbitrary object orientations, and densely packed targets with high visual similarity. These complexities demand highly efficient and discriminative interactive systems.

To address these challenges, current approaches to RSVG have broadly evolved into two primary paradigms: specialized supervised models and multimodal large language model (MLLM)-based methods. Specialized models [11] typically adopt a dedicated feature-alignment pipeline. They employ robust pretrained visual backbones (.g, ViT [12] and Swin-T [13]) and language encoders (e.g., BERT [14] and LSTM [15]) to independently extract unimodal representations, which are subsequently integrated using complex cross-modal fusion modules. More recently, MLLM-based methods [16], [17] have gained significant traction by adapting general-purpose foundation models (e.g., LLaVA [18], Qwen [19]) to the remote sensing domain. These advanced architectures leverage massive pretraining knowledge and instruction-tuning to perform reasoning-intensive grounding, often treating the localization task as a generative text or pixel-alignment problem. While both paradigms have achieved remarkable progress in bridging the semantic gap between language and aerial imagery, they share a fundamental operational constraint: they rely exclusively on explicit textual input.

Despite their architectural advancements, both paradigms share a fundamental bottleneck: a heavy reliance on explicit textual representation. In real-world operational scenarios, this dependency creates a critical barrier to deployment. As illustrated in Fig. 1(a), manual data entry is not only inefficient but often impractical for field operators who require hands-free interaction during time-critical remote sensing missions. Furthermore, reducing human intent to pure text inevitably strips away rich prosodic information, including tonal stress, prominence, and urgency. Such prosodic cues are known to play an important role in spoken-language comprehension, including lexical recognition, ambiguity resolution, and discourse interpretation [20], [21]. In remote sensing grounding, these acoustic cues may help discern subtle user intent, for example, distinguishing whether an operator is emphasizing the attribute "red" or the category "ship" when trying to disambiguate densely packed targets. While one might adopt automatic speech recognition (ASR) [22], [23] to enable voice input, this cascaded pipeline introduces a new information bottleneck. Prior studies on spoken language understanding

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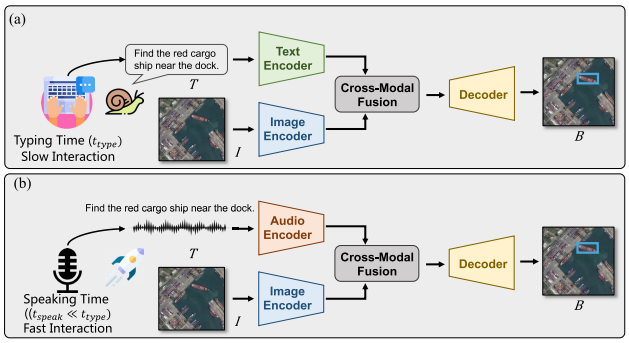



Fig. 1. Comparison between the previous method and our proposed audio-based RSVG frameworks. (a) Traditional text-based methods suffer from high interaction latency due to typing. (b) Our paradigm enables direct speech input, significantly reducing interaction time while preserving prosodic intent.

have shown that cascaded ASR-based systems are vulnerable to transcription errors, information loss, additional cost, and latency, motivating end-to-end speech understanding models that directly map acoustic inputs to semantic outputs [24], [25], [26]. This issue becomes more severe in specialized domains, where ASR systems may struggle with rare terms, jargon, or domain-specific vocabulary [27]. Consequently, minor transcription errors, such as misinterpreting "track" for "truck," can accumulate and cause catastrophic localization failures in downstream grounding.

To break these barriers, we advocate for a paradigm shift from text-dependent pipelines to direct audio-driven interaction, offering a far more intuitive and operational interface for time-critical remote sensing. We posit that an ideal interactive system should comprehend acoustic instructions natively, entirely bypassing the redundancy and error propagation of intermediate text generation. To this end, we formally introduce a novel task: audio-driven RSVG (Audio-RSVG) [as illustrated in Fig. 1(b)]. To support this research, we construct Audio-detectIon in optical remote sensing (DIOR), a comprehensive dataset featuring diverse acoustic variations and simulated environmental noise, together with a held-out test-acoustic split resynthesized using unseen voice styles and noise clips to evaluate acoustic generalization. Furthermore, we propose RSTalker, a streamlined, end-to-end framework specifically designed to directly map raw speech to visual targets. By eliminating cascaded ASR modules and heavy autoregressive text decoders, RSTalker avoids transcription bottlenecks. Instead, it employs a novel prosody-aware query generator to dynamically distill variable-length speech signals into compact, intent-conditioned semantic queries. These queries are then seamlessly integrated with hierarchical visual features to directly predict target coordinates. This direct audio-visual alignment not only drastically reduces the complete human–machine interaction latency, but also empowers the model to leverage nuanced acoustic emphasis (e.g., stressing "red" versus "ship") for intent-aware, disambiguated localization under complex conditions.

The main contributions are summarized as follows.

1) We propose the Audio-RSVG task for hands-free interaction. To support this, we construct the Audio-DIOR dataset, featuring a large-scale synthesized training set and a held-out test-acoustic split with unseen voice styles and noise conditions to evaluate acoustic generalization.
2) We develop RSTalker, an end-to-end framework that bypasses ASR systems to prevent error propagation. It features a prosody-aware query generator to directly align fine-grained acoustic cues with dense spatial visual features without intermediate text generation.
3) Experiments demonstrate RSTalker achieves state-of-the-art accuracy with strong robustness against environmental noise and phonetic ambiguity. By eliminating manual data entry and autoregressive text decoding, it delivers a 3.5× speedup in complete interaction-to-inference time over text-based baselines.

The remainder of this article is organized as follows. Section II reviews related work on remote sensing multimodal models, RSVG, and audio-visual multimodal learning. Section III formally defines the Audio-RSVG task, describes the construction of the Audio-DIOR dataset, and presents the proposed RSTalker framework. Section IV reports the experimental settings, compares RSTalker with existing grounding methods and multimodal foundation models, and provides ablation, robustness, and qualitative analyses. Finally, Section V concludes this article and discusses future research directions.

## II. Related Work

### A. Remote Sensing Multimodal Models

The paradigm of remote sensing image understanding has gradually evolved from unimodal visual representation learning to multimodal modeling, with the primary goal of bridging the pixel–semantic gap [4], [28]. Early efforts mainly focused on aligning remote sensing imagery with textual supervision. For example, Remotecontrastive language-image pre-training (CLIP) [29] adapted CLIP to the remote sensing domain for zero-shot recognition, while GRAFT [30] incorporated geo-location priors to improve vision–language representation learning.

With the rise of large language models, recent studies have further extended this line toward instruction-following multimodal systems. RSGPT [31] and EarthGPT [32] introduced LLM-based interaction capabilities for remote sensing understanding, while SkySense [33], [34] advanced large-scale multimodal pretraining with factorized encoders and broad multisensor coverage. More recently, GeoChat [16] and Falcon [35] pushed remote sensing multimodal models toward fine-grained grounding and unified reasoning across image-, region-, and pixel-level outputs.

Despite these advances, existing remote sensing multimodal research remains centered on the visual-language paradigm. Even when multiple sensors are considered, such as optical, SAR, or infrared imagery, the interaction channel is still predominantly text-based. As a result, audio has rarely been explored as an input modality for remote sensing understanding, especially for direct human–machine interaction and grounding.

### B. Remote Sensing Visual Grounding

RSVG aims to localize objects referred to by natural language expressions in remote sensing imagery [6].

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Compared with grounding in natural images, RSVG is particularly challenging due to extreme scale variation, arbitrary object orientation, and densely distributed targets. To address these challenges, earlier pipelines gradually evolved from two-stage designs to one-stage Transformer-based architectures, such as TransVG [36] and MGVLF [6], which rely on cross-modal attention to align textual queries with visual features. More recently, MLLM-based approaches have been introduced to improve generalization and reasoning ability. For instance, GeoGround [11] unifies horizontal bounding boxes (HBBs), oriented bounding boxes (OBBs), and segmentation tasks through text-style sequence modeling, while GeoViS [37] uses a VisualRAG-style agent to handle grounding in large-scale geospatial scenes. In addition, training-free methods, such as RSVG-ZeroOV [38] explore zero-shot open-vocabulary grounding with diffusion priors and attention evolution.

However, almost all existing RSVG methods assume that the user's intent is provided in explicit textual form. This assumption makes interaction dependent on manual typing, which is inefficient in time-sensitive scenarios and unsuitable for hands-free use. More importantly, once speech is converted into text, prosodic information, such as emphasis, urgency, and speaking style is inevitably discarded. Therefore, current RSVG research still treats grounding as a text-conditioned problem, leaving direct speech-conditioned grounding in remote sensing largely unexplored.

### C. Audio-Visual Multimodal Learning

Recent advances in multimodal learning [39], [40] have significantly benefited from bridging audio and visual modalities. Foundational models have been extended to handle diverse audio inputs; for instance, AudioCLIP [41] integrates the ESResNeXt audio model into the CLIP framework for zero-shot environmental sound classification. Similarly, SpeechCLIP [42] aligns speech and text through images, facilitating zero-shot speech–text retrieval without relying on transcriptions. Building upon robust representations, AVE-CLIP [43] proposes a temporal transformer for audio-visual event (AVE) localization in videos. In the realm of dense prediction, audio-visual segmentation (AVS) has emerged to generate pixel-level masks for sounding objects. Specifically, AVS [44] formalizes this task with the AVSBench dataset across various scenarios. To tackle its dynamic nature, AVSegFormer [45] leverages a dense audio-visual mixer and a sparse decoder to implicitly separate audio sources, while SAMA-AVS [46] adapts the pretrained segment anything model (SAM) to the AVS task using an annotation-free pipeline. Recent studies have explored spoken language as an explicit visual referring cue. Wnet [47] investigates audio-guided video object segmentation using spoken referring expressions, while STBridge [48] further studies noise-tolerant speech-referring video object segmentation. Different from these works, which mainly focus on object segmentation in natural-scene videos, our Audio-RSVG task addresses direct speech-conditioned object localization in static remote sensing imagery.

Crucially, our Audio-RSVG task differs fundamentally from these existing audio-visual works. While the aforementioned methods treat audio as intrinsic environmental sounds emitted by the objects themselves (e.g., a dog barking in a video), our task utilizes human spoken instructions as active queries to localize entirely silent targets in static remote sensing imagery. Therefore, rather than separating physical acoustic sources, our framework focuses on leveraging human prosodic cues to disambiguate densely packed geospatial targets, presenting a completely different task paradigm.

## III. Methodology

### A. Problem Formulation

Given a high-resolution remote sensing image $I \in \mathbb{R}^{H\times W\times 3}$ and a raw speech instruction $A = \{a_1, \ldots, a_L\}$ of time length $L$, the goal of Audio-RSVG is to predict the bounding box $B = (x, y, w, h)$ of the target object. Conventionally, RSVG methods rely on textual queries $T$, modeling the probability $P(B|I, T)$. However, they are hindered by slow typing interaction and lack of prosodic emphasis (e.g., stressing "red" versus "ship"), so we formulate this task as an end-to-end mapping. Our objective is to learn the parameters $\theta$ that directly maximize the probability of the ground-truth box given the multimodal inputs

$$B^* = \underset{B}{\operatorname{argmax}} P\left(B \middle| I, A; \theta\right). \tag{1}$$

This formulation bypasses explicit textual representation, requiring the model to learn a direct alignment between spatial visual features and temporal acoustic semantics.

### B. Audio-DIOR Dataset

Due to the lack of large-scale paired speech–image grounding data in remote sensing, we construct Audio-DIOR through an automated and scalable synthesis pipeline built upon an existing text-based benchmark Fig. 2. Let the source dataset be denoted as

$$D_{\text{src}} = \{(I_i, T_i, B_i)\}_{i=1}^{M} \tag{2}$$

where $I_i$ is a remote sensing image, $T_i$ is its referring expression, and $B_i$ is the corresponding ground-truth bounding box. Our objective is to transform each text query into multiple acoustically diverse spoken commands while preserving the original visual target and annotation. In this way, large-scale audio–visual grounding pairs can be obtained without reannotating images.

Before speech synthesis, each textual query is normalized to reduce pronunciation ambiguity caused by inconsistent writing forms. Specifically, we retain the semantic components that are most relevant to grounding, including target category words, attribute words, quantity expressions, and spatial relation phrases, while standardizing punctuation, abbreviations, numerals, and other irregular symbols. This preprocessing step improves synthesis stability and avoids introducing unnecessary acoustic variation unrelated to the grounding objective.

To generate speech queries, we replace the previous fixed-speaker TTS pipeline with Qwen3-TTS-12Hz-1.7B-VoiceDesign. Instead of synthesizing each sentence with a small set of predefined speakers, we exploit the voice-design and fine-grained control capabilities of Qwen3-TTS to produce richer yet controllable acoustic variations. Specifically, we

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define ten distinct timbre profiles using predefined natural-language prompts supported by Qwen3-TTS VoiceDesign. For each textual query $T_i$, three distinct timbre profiles are randomly selected from the profile bank to generate three acoustically diverse speech variants. Each selected profile specifies the desired acoustic characteristics, including timbre, speaking rate, speaking energy, articulation clarity, and speaking style. The synthesized speech is manually verified to ensure intelligibility and consistency with the intended voice characteristics. Accordingly, the number of generated speech variants for each query is $K = 3$. For each text query $T_i$, we construct a control instruction $p_k$ in natural language to describe the desired voice characteristics, such as timbre, speaking rate, speaking energy, articulation clarity, and speaking style, and then synthesize the corresponding audio as

$$A_i^{(k)} = G_{\text{Qwen}}(T_i; p_k), \quad k = 1, \ldots, K \tag{3}$$

where $G_{\text{Qwen}}(\cdot)$ denotes the Qwen3-TTS synthesis function and $K$ is the number of generated variants for each query. Compared with template-based multispeaker synthesis, this strategy allows us to model a broader range of speaking conditions while keeping the linguistic content unchanged.

We choose the 12 Hz variant rather than a conventional single-codebook TTS system for two reasons. First, the 12 Hz tokenizer explicitly disentangles semantic content from acoustic detail, allowing the generated speech to preserve both lexical information and controllable acoustic variation. Second, its multicodebook design is better suited to synthesizing expressive yet stable speech signals, which is beneficial when constructing a training corpus that aims to study direct audio–visual grounding rather than only speech recognition. Although streaming efficiency is not the primary focus of offline dataset construction, the low-latency and lightweight decoding properties of the 12 Hz model also make large-scale synthesis practically feasible.

To simulate deployment conditions in remote sensing scenarios, we further augment each clean synthetic utterance with environmental noise. Given a synthesized audio sample $A_i^{(k)}$, we sample an environmental noise clip $N$ and obtain the noisy variant

$$\tilde{A}_i^{(k)} = A_i^{(k)} + \lambda N \tag{4}$$

where $\lambda$ controls the noise level. In our implementation, the noise scaling coefficient $\lambda$ is randomly sampled from $[0, 0.4]$, corresponding to a noise augmentation strength ranging from 0% to 40%. This randomized noise strength exposes the model to acoustic interference of varying severity rather than a single fixed noise condition. The noise library contains representative operational sounds, such as wind noise, engine noise, mechanical noise, and other background interferences that may occur in outdoor or mission-oriented settings. By exposing the model to both clean and noisy speech during training, the resulting dataset supports robustness evaluation under more realistic acoustic conditions.

After controllable synthesis and noise augmentation, the final Audio-DIOR dataset is organized as

$$D_{\text{aud}} = \left\{ \left( I_i, \tilde{A}_i^{(k)}, B_i \right) \right\}_{i=1,k=1}^{M,K}. \tag{5}$$

TABLE I

STATISTICS OF THE AUDIO-DIOR DATASET. THE NUMBER OF IMAGES IS ESTIMATED FROM THE DIOR-RSVG SPLIT RATIO. EACH REFERRING EXPRESSION IS SYNTHESIZED INTO THREE SPEECH VARIANTS WITH DISTINCT TIMBRE PROFILES, RESULTING IN AN AVERAGE DURATION OF ABOUT 3.1 S PER AUDIO COMMAND

| Split | # Images | # Queries | Avg Time (s) | Total Time (h) |
|---|---|---|---|---|
| Train | 12,340 | 26,991 | 3.10 | 23.24 |
| Val | 1,690 | 3,829 | 3.08 | 3.28 |
| Test-Clean | 3,372 | 7,500 | 3.09 | 6.44 |
| Test-Acoustic | 3,372 | 7,500 | 3.09 | 6.44 |

Importantly, all synthetic variants derived from the same source triplet $(I_i, T_i, B_i)$ are assigned to the same data split to avoid information leakage across the training, validation, and test sets. Therefore, the evaluation protocol measures generalization to unseen image–query pairs rather than memorization of acoustically perturbed duplicates.

The statistics of Audio-DIOR are summarized in Table I. Following the original DIOR-RSVG data split, Audio-DIOR contains 26 991 training queries, 3829 validation queries, and 7500 test queries. Each referring expression is synthesized into one speech clip, resulting in an average duration of about 3.1 s per audio command. The training split contains 23.24 h of speech, while the validation and clean test splits contain 3.28 and 6.44 h, respectively. To evaluate acoustic generalization without introducing additional manual annotation, we further construct a test-acoustic split by resynthesizing the same test image–query pairs with held-out voice styles and unseen noise clips. This split has the same number of images and queries as test-clean, but provides a more challenging acoustic condition for robustness evaluation.

### C. RSTalker

We redesign RSTalker as a *hierarchical speech-conditioned grounding framework* Fig. 3. The architecture is motivated by two considerations. First, remote sensing images naturally exhibit multiscale structure, and the speech branch should provide stage-specific guidance rather than reusing the same query tokens across all visual stages. Second, an end-to-end audio grounding model should explicitly specify how speech features are converted into localization-oriented queries, how these queries interact with the visual hierarchy, and how the final bounding box is predicted and optimized. Accordingly, the RSTalker consists of four components: a visual encoder, a speech encoder, a speech-conditioned hierarchical query generator (SHQG), and a progressive cross-modal grounding head.

*1) Visual and Speech Encoders: Visual Encoder:* Given an input remote sensing image $I \in \mathbb{R}^{H \times W \times 3}$, we adopt a Swin Transformer as the visual backbone to extract a hierarchical feature pyramid

$$\left\{ F_v^{(1)}, F_v^{(2)}, F_v^{(3)}, F_v^{(4)} \right\} = E_v(I) \tag{6}$$

where $F_v^{(s)} \in \mathbb{R}^{H_s W_s \times d}$ denotes the flattened visual feature map at stage $s$, and $d$ is the hidden dimension after projection. This hierarchical representation is well suited to remote sensing

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imagery, where targets may vary drastically in size, orientation, and spatial density.

*Speech Encoder:* Given a speech command $A$, we first convert the raw waveform into a log-mel spectrogram and feed it into a pretrained Whisper encoder. Following the original design philosophy of bypassing transcription, we retain only the encoder branch and discard the autoregressive text decoder. The speech encoder outputs frame-level acoustic representations

$$F_a^{(0)} = E_a(A) \in \mathbb{R}^{L\times d} \tag{7}$$

where $L$ is the temporal length of the encoded sequence. To align the speech branch with the hierarchical visual backbone, we further construct a temporal speech pyramid by progressively aggregating $F_a^{(0)}$ with lightweight temporal convolution and strided downsampling

$$F_a^{(s)} = \phi^{(s)}\left(F_a^{(s-1)}\right), \quad s = 1,2,3,4 \tag{8}$$

where $\phi^{(s)}(\cdot)$ denotes a 1-D convolutional aggregation block. In this way, the speech branch provides stage-specific representations with different temporal receptive fields, rather than a single shared query set for all visual stages.

*2) Speech-Conditioned Hierarchical Query Generator:* Directly using frame-level speech features for grounding is suboptimal because the sequence is variable in length and contains substantial redundancy, such as silence, hesitation, and nondiscriminative phonetic fragments. To address this issue, we introduce a SHQG, which converts each level of the speech pyramid into a compact set of localization-oriented query tokens.

For each stage $s$, we first enhance the saliency of the speech representation through a gated temporal projection

$$\tilde{F}_a^{(s)} = \sigma\left(W_g^{(s)}F_a^{(s)}\right) \odot W_p^{(s)}F_a^{(s)} \tag{9}$$

where $W_g^{(s)}$ and $W_p^{(s)}$ are learnable linear projections, $\sigma(\cdot)$ is the sigmoid activation, and $\odot$ denotes element-wise multiplication. The gating operation suppresses temporally redundant regions and highlights acoustically informative segments.

We then map the variable-length sequence $\tilde{F}_a^{(s)}$ to a fixed number of query tokens through adaptive temporal pooling

$$Q_a^{(s)} = \mathrm{MLP}^{(s)}\left(\mathrm{Pool}\left(\tilde{F}_a^{(s)}, N_s\right)\right) \tag{10}$$

where $Q_a^{(s)} \in \mathbb{R}^{N_s\times d}$ and $N_s$ denotes the number of speech queries used at stage $s$. Unlike static object queries, $Q_a^{(s)}$ is dynamically generated from the input speech and varies across stages. This design allows coarse stages to focus on global command semantics while finer stages retain more localized speech-side cues related to attributes or spatial relations.

*3) Progressive Cross-Modal Alignment:* Given the stage-specific speech queries $\{Q_a^{(s)}\}_{s=1}^4$ and the visual pyramid $\{F_v^{(s)}\}_{s=1}^4$, we perform progressive cross-modal alignment from coarse to fine. At each stage, the decoder contains three sublayers: self-attention over speech queries, cross-attention from speech queries to visual features, and a feed-forward network.

For the first stage, the decoder input is directly initialized by the speech queries

$$H^{(1)} = \mathcal{D}^{(1)}\left(Q_a^{(1)}, F_v^{(1)}\right). \tag{11}$$

For deeper stages, the hidden state from the previous stage is first projected to the current query space and then fused with the current-stage speech queries

$$H^{(s)} = \mathcal{D}^{(s)}\left(Q_a^{(s)} + \psi^{(s)}\left(H^{(s-1)}\right), F_v^{(s)}\right), \quad s = 2,3,4 \tag{12}$$

where $\psi^{(s)}(\cdot)$ denotes a linear projection or upsampling adapter. This progressive formulation enables information flow across stages and avoids the degenerate design in which the same speech tokens are repeatedly copied into all visual layers.

Inside each decoder stage, the query update can be written as

$$\hat{H}^{(s)} = \mathrm{MHSA}\left(H^{(s)}\right) \tag{13}$$

$$\bar{H}^{(s)} = \mathrm{MHCA}\left(\hat{H}^{(s)}, F_v^{(s)}, F_v^{(s)}\right) \tag{14}$$

$$H_{\mathrm{out}}^{(s)} = \mathrm{FFN}\left(\bar{H}^{(s)}\right) \tag{15}$$

where MHSA and MHCA denote multihead self-attention and multihead cross-attention, respectively. The attention weights are preserved during inference and can be visualized to provide interpretable evidence of speech-to-region alignment.

*4) Query-Guided Grounding Head:* After the final alignment stage, we obtain the refined speech-conditioned query set $H_{\mathrm{out}}^{(4)} \in \mathbb{R}^{N_4\times d}$. We aggregate these query tokens into a compact grounding representation through attention pooling

$$\alpha_n = \frac{\exp\left(w^\top h_n\right)}{\sum_{m=1}^{N_4}\exp\left(w^\top h_m\right)}, \qquad z = \sum_{n=1}^{N_4}\alpha_n h_n \tag{16}$$

where $h_n$ is the $n$th query token and $w$ is a learnable vector. The aggregated representation $z$ is then passed to a lightweight regression head to predict the normalized bounding box

$$\hat{B} = \sigma\left(\mathrm{MLP}_{\mathrm{box}}(z)\right) = \left(\hat{c}_x, \hat{c}_y, \hat{w}, \hat{h}\right) \tag{17}$$

where $(\hat{c}_x, \hat{c}_y, \hat{w}, \hat{h}) \in [0,1]^4$ denote the center coordinates, width, and height of the predicted box in normalized image coordinates.

To stabilize optimization, we additionally attach auxiliary box heads to intermediate decoder outputs and apply deep supervision

$$\hat{B}^{(s)} = \sigma\left(\mathrm{MLP}_{\mathrm{aux}}^{(s)}\left(\mathrm{Pool}\left(H_{\mathrm{out}}^{(s)}\right)\right)\right), \quad s = 1,2,3. \tag{18}$$

*5) Training Objective:* Since each image–speech pair corresponds to a single target instance, we optimize RSTalker with a direct box regression objective composed of an $\ell_1$ loss and a generalized IoU loss

$$\mathcal{L}_{\mathrm{main}} = \lambda_1\left\|\hat{B} - B\right\|_1 + \lambda_{\mathrm{giou}}\mathcal{L}_{\mathrm{GIoU}}\left(\hat{B}, B\right) \tag{19}$$

where $B$ is the ground-truth box. The auxiliary stage-wise losses are defined in the same form

$$\mathcal{L}_{\mathrm{aux}} = \sum_{s=1}^{3}\gamma_s\left[\lambda_1\left\|\hat{B}^{(s)} - B\right\|_1 + \lambda_{\mathrm{giou}}\mathcal{L}_{\mathrm{GIoU}}\left(\hat{B}^{(s)}, B\right)\right]. \tag{20}$$

The overall objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{main}} + \mathcal{L}_{\mathrm{aux}}. \tag{21}$$

The resulting framework preserves the end-to-end audio-to-grounding paradigm while introducing stage-specific speech

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guidance, a complete grounding head, and explicit optimization targets. Compared with the initial formulation, the design provides a more faithful coarse-to-fine alignment mechanism between temporal speech cues and hierarchical visual features.

**Algorithm 1** Inference Procedure of RSTalker

**Require:** Remote sensing image $I$; speech command $A$; trained RSTalker.

**Ensure:** Predicted bounding box $\hat{B}$.

```
# Extract hierarchical visual features.
{F_v^(1), F_v^(2), F_v^(3), F_v^(4)} = E_v(I)
# Encode speech into frame-level acoustic features.
F_a^(0) = E_a(A)
# Build temporal speech pyramid and stage-specific queries.
for s = 1 to 4 do
    F_a^(s) = φ^(s)(F_a^(s−1))
    Q_a^(s) = Q^(s)(F_a^(s))
end for
# Initialize the first cross-modal alignment stage.
H^(1) = D^(1)(Q_a^(1), F_v^(1))
# Progressively refine speech-conditioned visual representations.
for s = 2 to 4 do
    H^(s) = D^(s)(Q_a^(s) + ψ^(s)(H^(s−1)), F_v^(s))
end for
# Aggregate final queries and predict the target box.
z = AttnPool(H^(4))
B̂ = σ(MLP_box(z))
return B̂.
```

*6) Inference:* To make the test-time pipeline explicit, Algorithm 1 summarizes the forward procedure of RSTalker. Given a remote sensing image and a speech command, RSTalker first extracts a visual feature pyramid and a frame-level acoustic representation. The acoustic representation is further transformed into a temporal speech pyramid, from which SHQG generates stage-specific speech queries. These queries are progressively aligned with multiscale visual features, and the final query-guided grounding head predicts the target bounding box.

Compared with a cascaded audio → text system, the inference path of RSTalker is also more direct. The model does not need to first decode speech into a discrete text sequence and then feed it into a separate grounding model. Instead, the acoustic representation is continuously propagated through the speech pyramid and progressively fused with the visual hierarchy. This avoids potential transcription errors and keeps the audio-to-grounding process fully end-to-end.

## IV. Experiments

### A. Evaluation Setup

*1) Baselines and Metrics:* We compare RSTalker with both specialized grounding models and recent remote sensing multimodal foundation models on the DIOR-RSVG dataset. The specialized baselines include ZSGNet [49], FAOA [52], ResC [54], LBYL-Net [55], TransVG [56], VLTVG [57], MGVLF [38], and RemoteSAM [58], while the foundation-model baselines include SkyEyeGPT [59], LHRS-Bot [62], GeoGround [11], Falcon [35], and GeoZero [66]. In addition to this broad comparison, we conduct controlled modality analysis with three input settings: text, audio → text, and audio-only. The text setting uses the original referring expression; the audio → text setting first transcribes speech into text through ASR [23] and then performs grounding; and the audio-only setting directly predicts the target box from speech without intermediate transcription.

We use mean intersection-over-union (mIoU) and Acc@0.5 as the main evaluation metrics, following standard RSVG evaluation protocols [6]. mIoU measures the average overlap between predicted and ground-truth boxes, while Acc@0.5 reports the percentage of samples whose IoU exceeds 0.5. We also report model parameters, inference time, user input time, and total interaction time to evaluate the efficiency of different input paradigms.

*2) Training Details:* We implement RSTalker using PyTorch and conduct all experiments on a workstation equipped with two NVIDIA GeForce RTX 4090 GPUs. For the model architecture, we employ Swin-T [13] pretrained on ImageNet-22K [70] as the visual backbone, resizing input images to 800 × 800 to preserve fine-grained details typical in remote sensing. The audio branch is initialized with the encoder of Whisper-Base.en [23] to leverage robust acoustic representations. We evaluate performance on the proposed Audio-DIOR dataset using *Acc*@0.50 and mIoU metrics. The model is optimized end-to-end using the AdamW optimizer with a weight decay of $10^{-4}$ for 50 epochs. Due to the high resolution of remote sensing imagery, we set the total batch size to 4. Both pretrained encoders are kept frozen throughout training, and only the task-specific modules for hierarchical speech query generation, progressive cross-modal alignment, and bounding-box prediction are optimized.

### B. Main Results

*1) Comparison on Accuracy:* Table II compares RSTalker with both specialized RSVG methods and recent multimodal foundation models. Overall, RSTalker achieves the best *Acc*@0.5 among all compared methods, reaching 79.85%, while also obtaining the second-best *mIoU* at 78.29%. Notably, this result is achieved with only 126 M parameters, indicating that direct speech-conditioned grounding can remain highly competitive without relying on billion-scale models.

Compared with specialized grounding methods, RSTalker establishes a clear advantage in localization accuracy. In particular, it improves *Acc*@0.5 over MGVLF by 3.07 points, while maintaining nearly the same *mIoU*. This result suggests that the proposed framework is especially effective at resolving fine-grained ambiguities in remote sensing scenes, where targets are often small, densely distributed, and visually similar.

Compared with recent multimodal foundation models, RSTalker also shows a strong accuracy–efficiency tradeoff. Although these models benefit from much larger pretrained backbones and substantially higher parameter counts, our method still delivers the best reported *Acc*@0.5 in this

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TABLE II

BROAD COMPARISON WITH EXISTING RSVG METHODS AND MULTIMODAL FOUNDATION MODELS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. "-" INDICATES THAT THE METRIC WAS NOT REPORTED

| Methods | Publication | Image Encoder | Text/Audio Encoder | Parameters | *mIoU* | *Acc*@0.5 |
|---|---|---|---|---|---|---|
| *Specialized Models* | | | | | | |
| ZSGNet [49] | ICCV19 | ResNet-50 [50] | Bi-LSTM [51] | 140M | 44.12 | 51.67 |
| FAOA [52] | CVPR20 | DarkNet-53 [53] | LSTM [51] | 150M | 62.86 | 70.86 |
| ResC [54] | CVPR21 | DarkNet-53 [53] | BERT [14] | 150M | 64.24 | 72.71 |
| LBYL-Net [55] | TIP22 | DarkNet-53 [53] | BERT [14] | 155M | 65.92 | 73.78 |
| TransVG [56] | NeurIPS21 | ResNet-101 [50] | BERT [14] | 136M | 76.27 | 72.41 |
| VLTVG [57] | CVPR22 | ResNet-101 [50] | BERT [14] | 155M | 77.85 | 75.79 |
| MGVLF [38] | TGRS23 | ResNet-101 [50] | BERT [14] | 136M | **78.42** | 76.78 |
| RemoteSAM [58] | ACM MM25 | Swin-T [13] | BERT [14] | 180M | 65.07 | 74.36 |
| *Multi-modal Foundation Models* | | | | | | |
| SkyEyeGPT [59] | NeurIPS22 | EVA-CLIP ViT-G [60] | LLaMA2 [61] | 7B | - | 70.5 |
| LHRS-Bot [62] | ECCV24 | CLIP ViT-L/14 [63] | LLaMA2 [61] | 7B | - | 73.45 |
| GeoGround [11] | arXiv24 | CLIP ViT-L/14 [63] | LLaVA-1.5-7B [64] | 7B | - | 77.73 |
| Falcon [35] | arXiv25 | Florence-2-0.7B [65] | Florence-2-0.7B [65] | 0.7B | - | 56.87 |
| GeoZero [66] | arXiv25 | SigLip 2 [67] | Qwen3 [68] | 8B | - | <u>79.63</u> |
| **Ours** | - | Swin-T [13] | Whisper [69] | **126M** | <u>78.29</u> | **79.85** |

TABLE III

EFFICIENCY COMPARISON OF DIFFERENT INPUT MODALITIES. "INPUT TIME" MEASURES THE AVERAGE TIME FOR A USER TO COMPLETE AN INSTRUCTION (TYPING VERSUS SPEAKING). "PARAMETERS" FOR AUDIO2TEXT INCLUDE BOTH THE ASR AND THE GROUNDING MODEL

| Input Type | Input Time | Inference Time | Total Time | Parameters |
|---|---|---|---|---|
| Text | 18s | 30.45ms | 18.3s | 180M |
| Audio2Text | 10s | 30.45ms | 10.3s | 74M+180M |
| Audio | **5s** | **26.73ms** | **5.3s** | **126M** |

TABLE IV

ANALYSIS OF THE MODALITY GAP. THE FIRST ROW DENOTES USING TEXT INPUT. "AUDIO2TEXT" EMPLOYS A CASCADED ASR PIPELINE. "AUDIO-ONLY" IS OUR RSTALKER

| Audio2Text | Audio-only | mIoU | Acc@0.5 |
|---|---|---|---|
| | | **78.42** | 76.78 |
| ✓ | | 76.71 | 75.29 |
| | ✓ | 78.29 | **79.85** |

TABLE V

EFFECTIVENESS OF THE KEY MODULES IN THE RSTALKER

| Speech Pyramid | SHQG | Progressive Alignment | mIoU | Acc@0.5 |
|---|---|---|---|---|
| | | | 42.41 | 51.30 |
| ✓ | | | 65.89 | 72.42 |
| ✓ | ✓ | | 69.97 | 74.28 |
| ✓ | | ✓ | 71.48 | 75.61 |
| ✓ | ✓ | ✓ | **78.29** | **79.85** |

comparison. For example, it slightly exceeds GeoZero in *Acc*@0.5 despite using a far more compact architecture. This observation indicates that, for remote sensing grounding, a task-specific end-to-end audio–visual design can be more effective than simply scaling up a generic vision-language model.

*2) Comparison on Efficiency:* As shown in Table III, RSTalker demonstrates superior efficiency in both interaction speed and model complexity. While text-based methods suffer from a severe input bottleneck (18 s for typing) and cascaded Audio2Text approaches incur high parameter costs (combining ASR and grounding models for 254 M params), our approach drastically reduces the total interaction time to 5.3 s—a 3.5× speedup over manual entry. Furthermore, with only 126 M parameters, RSTalker is significantly lighter than cascaded baselines, making it the most viable solution for real-time, resource-constrained remote sensing applications.

### C. Ablation Studies

*1) Analysis of Different Input Modalities:* Table IV examines the gap between text-based grounding, cascaded speech-to-text grounding, and direct audio-driven grounding. When the textual query is replaced by an ASR transcript, performance drops on both mIoU and Acc@0.5. This result indicates that the cascaded pipeline is vulnerable to transcription errors, especially when remote sensing expressions contain fine-grained object categories, attributes, or spatial relations. Once these linguistic cues are corrupted by ASR, the downstream grounding model is easily misled.

In contrast, the direct audio-only setting achieves the best overall performance. It not only recovers the loss introduced by the ASR pipeline, but also surpasses the text-input baseline in Acc@0.5 while maintaining competitive mIoU. This result suggests that direct speech grounding is able to preserve useful acoustic information that is discarded in text conversion. In particular, prosodic cues and fine-grained speech variations may provide additional evidence for disambiguating visually similar targets in cluttered remote sensing scenes.

*2) Effectiveness on Key Modules of RSTalker:* We present the ablation results of the RSTalker in Table V. The direct audio grounding baseline performs poorly, indicating that naive audio–visual grounding is insufficient for this task. Introducing the speech pyramid brings a large improvement, which confirms the importance of constructing hierarchical speech representations that match the multiscale visual backbone. Adding SHQG yields further gains by transforming

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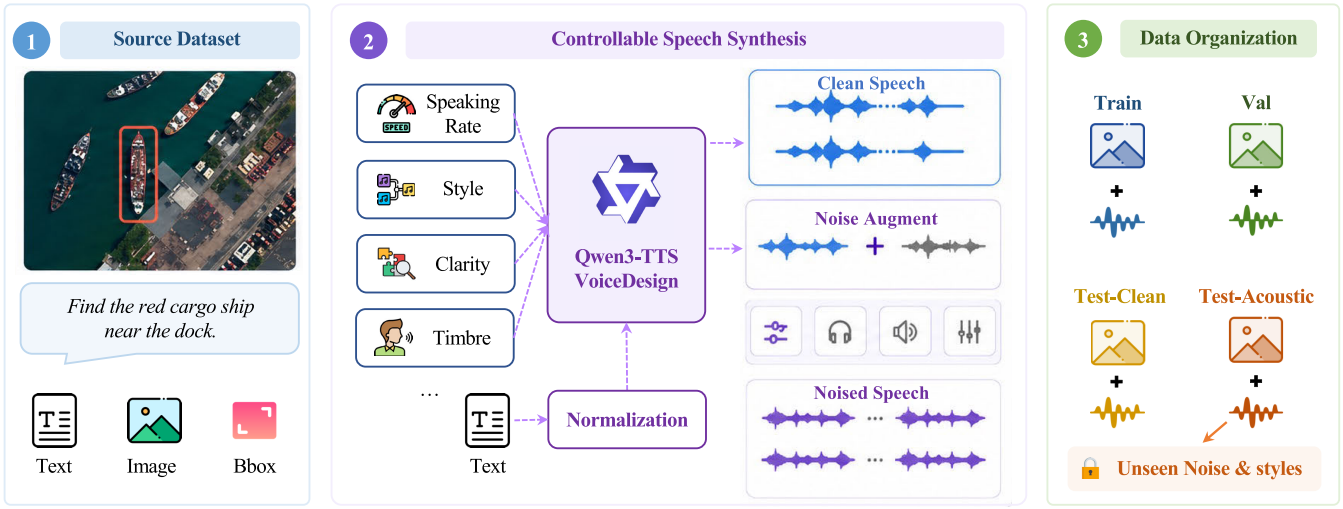



Fig. 2. Automated construction pipeline of Audio-DIOR. We convert the original DIOR-RSVG image–text-bbox triplets into audio-driven grounding samples through text normalization, controllable speech synthesis, and noise augmentation. Qwen3-TTS VoiceDesign is used to synthesize speech commands with diverse speaking rates, styles, clarity, and timbre, while environmental noises are injected to simulate challenging acoustic conditions. The resulting samples follow the original train/validation/test split, and an additional test-acoustic split is built with held-out voice styles and unseen noise clips to evaluate acoustic generalization.

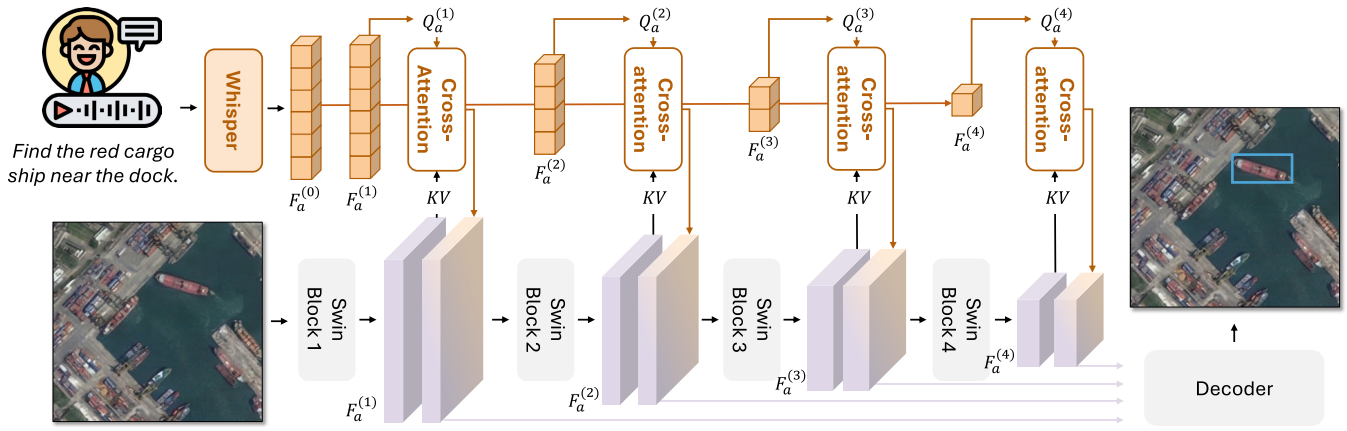



Fig. 3. Architecture of RSTalker. It leverages a Swin Transformer and a Whisper encoder to extract visual and acoustic representations, respectively. The core mechanism involves dynamic audio queries ($Q$) interacting with multiscale visual features ($KV$) through stage-wise cross-attention layers. This design facilitates deep multimodal fusion, allowing the model to ground objects directly from speech signals without intermediate text generation.

variable-length speech features into compact and localization-oriented query tokens. Enabling progressive alignment also consistently improves performance, showing that coarse-to-fine cross-modal refinement is more effective than independent stage-wise matching. Combining all three modules gives the best results, demonstrating that they play complementary roles in the final framework.

*3) Robustness Under Acoustic Difficulty:* To verify whether direct audio grounding is more robust than cascaded transcription under noisy environments, we evaluate both audio → text and RSTalker under clean speech, wind noise, engine noise, and mixed noise. As shown in Table VI, the audio → text baseline is highly sensitive to acoustic interference. Its Acc@0.5 drops from 75.29 under clean speech to 73.80 with wind noise and 71.53 with engine noise, and further decreases to 69.95 under mixed noise. This indicates that noise can corrupt the intermediate ASR transcript and propagate errors to the downstream grounding model.

By contrast, RSTalker maintains stable performance across all noisy conditions, achieving 78.09, 78.38, and 78.24 Acc@0.5 under wind, engine, and mixed noise, respectively. The small gap between clean and noisy settings suggests that direct audio grounding is less dependent on a brittle text-conversion step. These results support our design choice of directly aligning acoustic representations with visual features instead of relying on intermediate transcription.

*4) Effectiveness of Audio-DIOR Construction Strategies:* To examine whether the proposed Audio-DIOR construction pipeline improves acoustic generalization, we evaluate different training-data variants on a held-out acoustic test set. The test set uses the same images, referring expressions, and bounding-box annotations as the original test split, but the

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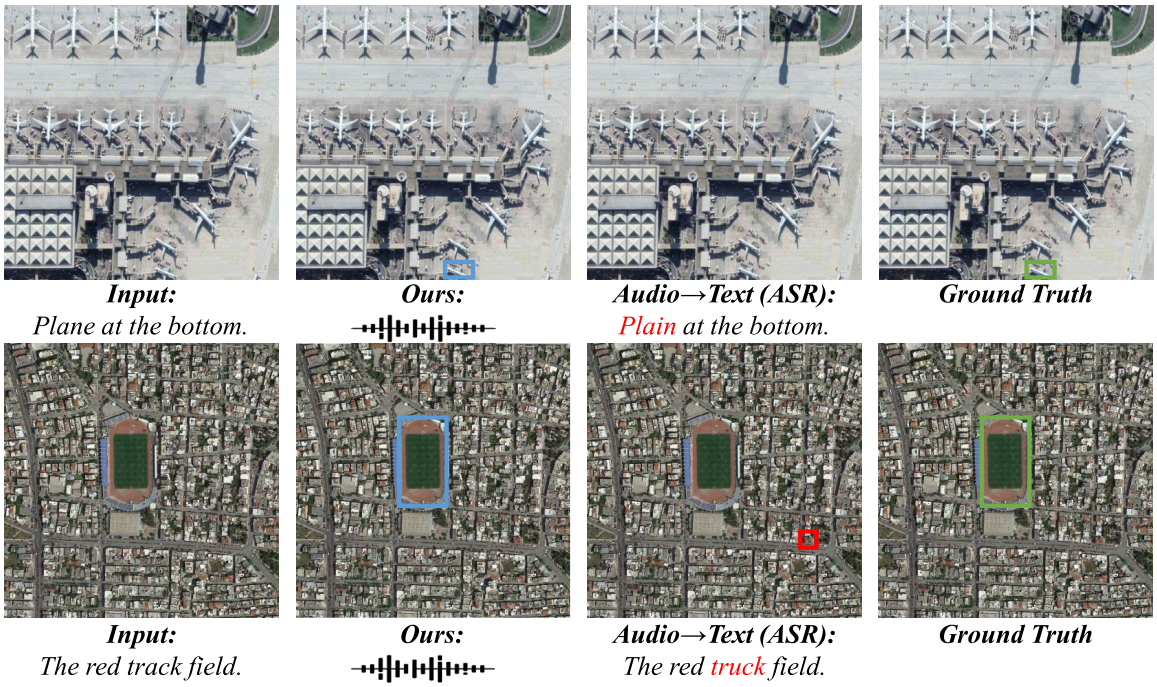



Fig. 4. Qualitative results of RSTalks and ASR-based RSVG methods. When ASR errors occur, it impacts model performance.

TABLE VI

ROBUSTNESS UNDER DIFFERENT ACOUSTIC CONDITIONS. WE REPORT ACC@0.5 FOR THE CASCADED AUDIO → TEXT PIPELINE AND THE PROPOSED RSTALKER

| Method | Clean | Wind | Engine | Mixed |
|---|---|---|---|---|
| Audio→Text | 75.29 | 73.80 | 71.53 | 69.95 |
| RSTalker | **79.85** | **78.09** | **78.38** | **78.24** |

TABLE VII

EFFECTIVENESS OF KEY AUDIO-DIOR CONSTRUCTION STRATEGIES UNDER HELD-OUT ACOUSTIC CONDITIONS. ALL VARIANTS ARE EVALUATED ON AN UNSEEN-ACOUSTIC TEST SET SYNTHESIZED WITH HELD-OUT VOICE STYLES AND UNSEEN NOISE CLIPS

| VoiceDesign | Multi-Variant | Noise Aug. | $mIoU$ | $Acc$@0.5 |
|---|---|---|---|---|
| | | | 62.71 | 64.59 |
| ✓ | | | 67.22 | 67.80 |
| ✓ | ✓ | | 71.53 | 72.64 |
| ✓ | ✓ | ✓ | **78.29** | **79.85** |

speech queries are resynthesized with unseen voice styles and mixed with unseen noise clips.

As shown in Table VII, the model trained with basic TTS data performs poorly under held-out acoustic conditions, achieving only 62.71 mIoU and 64.59 Acc@0.5. After introducing VoiceDesign, the performance improves to 67.22 mIoU and 67.80 Acc@0.5, indicating that controllable variations in timbre, speaking rate, energy, and clarity help the model adapt to unseen speech styles. Multivariant synthesis further improves the result to 71.53 mIoU and 72.64 Acc@0.5, showing that generating multiple speech realizations for the same query provides richer acoustic coverage and reduces overfitting to a single synthetic voice pattern. Finally, adding noise augmentation leads to the best performance, reaching 78.29 mIoU and 79.85 Acc@0.5. This substantial gain demonstrates that environmental noise simulation is essential for robust audio-driven grounding, especially when the test speech contains unseen acoustic interference. Overall, these results confirm that Audio-DIOR is not merely a speech-converted version of DIOR-RSVG; its controllable voice generation, multivariant synthesis, and noise augmentation jointly improve the model's robustness under realistic acoustic shifts.

### D. Qualitative Results

Fig. 4 highlights a critical advantage of our end-to-end paradigm: robustness against phonetic ambiguity. In scenarios where ASR fails due to phonetic similarities—such as misinterpreting "track" (stadium) as "truck" (vehicle) or "plane" as "plain"—the cascaded baseline is misled into localizing incorrect objects (e.g., mistaking a red building for the target vehicle). In contrast, RSTalker successfully bypasses this textual bottleneck by learning directly from acoustic features. It accurately grounds the target objects even when phonetic cues are ambiguous, effectively eliminating the error propagation inherent in text-based methods.

## V. CONCLUSION

In this work, we introduce Audio-RSVG, a new task that studies how to localize targets in remote sensing imagery directly from spoken instructions. To support this setting, we construct Audio-DIOR, a speech-augmented benchmark built from existing remote sensing grounding data with controllable acoustic variation and environmental noise. We further propose RSTalker, a hierarchical speech-conditioned grounding framework that combines a temporal speech pyramid, stage-specific

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query generation, and progressive cross-modal alignment to bridge speech cues with multiscale visual features. Experimental results show that the proposed framework achieves strong grounding performance and compares favorably with existing specialized models and large multimodal baselines, while also avoiding the error propagation introduced by cascaded ASR-based pipelines. We hope this work can encourage further research on more natural and efficient human–machine interaction for remote sensing understanding.

Despite the encouraging results, several limitations remain in the current study. First, Audio-DIOR is still predominantly constructed from synthesized speech. Although controllable voice generation and noise augmentation provide diverse acoustic conditions, broader evaluation with real human speech covering diverse accents, speaking styles, and spontaneous expressions is required to further assess real-world generalization. Second, the current Audio-RSVG formulation focuses on HBB grounding, while OBBs, segmentation, and more complex geospatial reasoning tasks remain to be explored. Third, the robustness and interpretability of the proposed framework can be investigated more systematically through harder unseen acoustic conditions, more challenging ambiguous referring expressions, and stage-wise attention analysis. Finally, lightweight deployment and multilingual spoken interaction remain important directions for improving the practical applicability of audio-driven remote sensing systems in real field environments.

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